Growth and spatial inequalities

Gabriel M. Ahlfeldt
Humboldt University (BSoE, LSE, CEPR)

Challenges and policy implications
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Aim & scope

- **Cities** are the powerhouses of the economy
  - We know this since Marshall (1890)
- Let's think about **why they are pulling (further) ahead**, recently?
  - What has changed?
  - What did we (or I) learn?
- Big-picture thoughts from **subjective reading of literature**
  - **Increasing returns to density** favor big cities
  - **Structural transformation** favors big cities
  - Increase in market concentration (**superstar firms**) favors big cities
  - **Urban QoL premium** increasingly favors big cities

Why do people live in cities?

- Perhaps the oldest question in urban economics...
  - What is good about **density = concentration of people and firms in space**?
- Evergreen answer: Cities are **good places to produce**
  - Agglomeration economies lead to higher productivity and wages, innovation and entrepreneurship...
- More recent answer: Cities are **good places to live**
  - Big-city consumption amenities and variety!
- Of course, there are also urban costs
  - housing costs, congestion, inequality

Q: Which benefits and costs dominate?

Benefits and costs of density

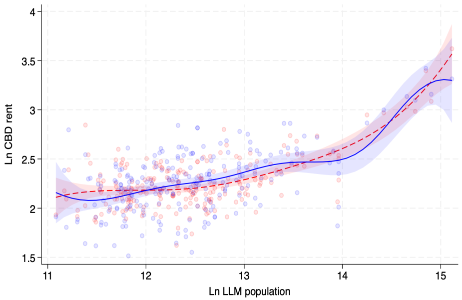
ID	Outcome	Elasticity	Quantity, p.c., year	Unit value	**PV of 1% dens. incr. ($)**
1	Wage	0.04	Income ($) 35,000	–	**280**
2	Patent intensity	0.21	Patents (#) 2.06E–04	Patent value ($/#) 793k	**7**
3	Rent	0.15	Income ($) 35,000	Expenditure share 0.33	**347**
4	VMT reduction	0.06	VMT (mile) 10,658	Private cost $/mile 0.83	**107**
5	Variety value	0.12	Income ($) 35,000	Expenditure share 0.14	**115**
6	Local public spending	0.17	Total spending ($) 1,463	–	**50**
7	Wage gap reduction	-0.035	Income ($) 35,000	Inequality premium 0.048	**-12**
8	Crime rate reduction	0.085	Crimes (#) 0.29	Full cost ($/#) 3,224	**16**
9	Green density	0.28	Green area (p.c., m^2) 540	Park value ($/$m^2$) 0.3	**100**
10	Pollution reduction	-0.13	Rent ($) 11,550	Rent–poll. elasticity 0.3	**-90**
11	Energy use reduction	0.07	Energy (1 M BTU) 121.85	Cost ($/1 M BTU) 18.7	**32**
		0.07	CO_2 emissions (t) 25	Social cost ($/t) 43	**15**
12	Average speed	-0.12	Driving time (h) 274	VOT ($/h) 10.75	**-71**
13	Car use reduction	0.05	VMT 10,658	Social cost ($/mile) 0.016	**2**
14	Health	-0.09	Mortality risk (#) 5.08E–04	Value of life ($/#) 7M	**-64**
15	Self-reported well-being	-0.004	Income ($) 35,000	Inc.–happ. elasticity 2	**-52**

Notes: Elasticities are taken from a quantitative literature review synthesising 347 estimates across 180 studies. Monetary values report long-run per-capita present values of a 1% increase in population density for a representative high-income metropolitan area. Source: Ahlfeldt & Pietrostefani (2019).

Urban productivity premium may be even larger

- **Higher wages** reflect greater productivity
  - **Sharing:** shared inputs, infrastructure, specialized suppliers
  - **Matching:** thicker labor markets, better job-worker matches
  - **Learning:** knowledge spillovers, peer effects, dense networks
- Probably understating the effect
  - City-size elasticity of commercial rent: 15%
  - Implied TFP effect: 2% ⇒ City-size **elasticity larger than when inferred from wages and rents**
- QoL matters too (more later)...

CBD rent vs city population



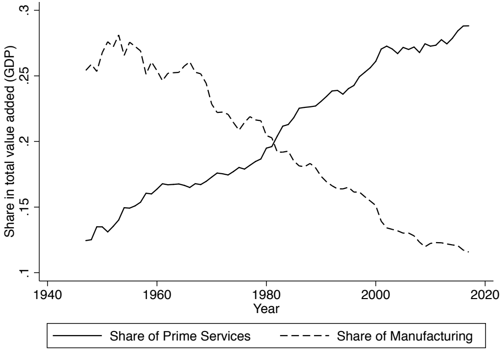
Source: Ahlfeldt, Heblich, Seidel, Yin (2026).

What is changing?

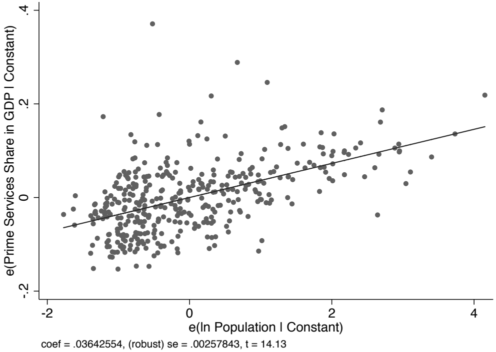
- Agglomeration economies explain spatial concentration
  - Production **and** consumption sides matter
- **But spatial divergence is increasing over time**
  - „Cities with the right industry mix and a solid human capital base attract ever more good firms and pay high wages. Cities at the other end of the spectrum are stuck with dead jobs and low wages." (Moretti, 2013)
  - Rich, urban „superstar" regions are growing faster (Iammarino et al., 2019)
  - Urban wage premium almost doubled between 1985 and 2015 (Dauth et al., 2022)
  - Commercial rent premium also increased (Ahlfeldt, Heblich, Seidel, Fan, 2026)

Q: What is driving the widening urban–rural gap?

Structural transformation in cities



(a) Share of prime services in US GDP



(b) Share of prime services and MSA size

Source: Ahlfeldt, Albers, Behrens (2022)

Structural transformation is urban-biased

- Structural transformation shifts activity toward **services** and **intangibles**
  - Manufacturing declines in relative terms
- The growing sectors are also the **most urban-complementary**
  - they **rely on dense labor markets, teams, and networks**
  - returns to density increase even without manufacturing decline
- Transformation is **not spatially neutral**:
  - large cities gain employment, productivity, and wages
  - smaller regions fall behind in relative terms
- Evidence
  - Chen, Novy, Perroni, Wong (2025) for France
  - Eckert, Ganapati and Walsh (2025) for US

The rise of the superstar firm

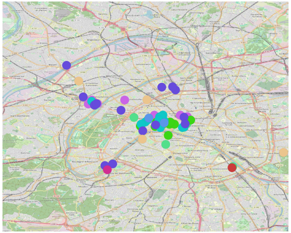
- There was not only a **shift** between sectors, also **within sectors**
  - Productivity growth has become increasingly **uneven across firms**
- A small set of highly productive **superstar firms** pull away from the rest
  - higher productivity growth at the frontier
  - rising market shares, markups, and profits
- **Technological change** and scale economies reinforce this pattern
  - **digital technologies favour scalability** and winner-take-most dynamics
- E.g. Autor et al. (2020); De Loecker et al. (2020)

Q: Do superstar firms contribute spatial concentration?

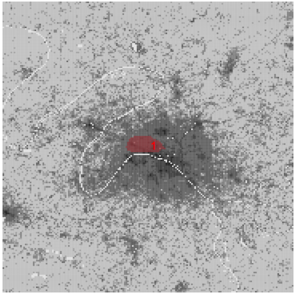
Where do superstar firms locate?

- **Superstar firms** concentrate in **superstar cities**
  - Large cities high-skilled labour force (Gourko, Mayer, Sinai, 2013).
- **Superstar firms** concentrate in **prime locations**
  - Clusters of extreme density and spillovers (Ahlfeldt, Albers, Behrens, 2026)
  - E.g. Midtown Manhattan: 1.7M on 11 km^2
- **Prime locations** only exist in **superstar cities**
  - Critical mass for professional and social networks
- **Rise of superstar firm benefits superstar cities**
  - Uneven productivity growth **across firms** leads to...
  - ...uneven economic growth **across regions**
    - 173 out of 278 new superstar firms in Ile-de-France (France, 1998-2016, CASD data)

Superstar firm openings since 2000



Prime location (Ahlfeldt, Albers, Behrens, 2026)



What drives the location of superstar firms?

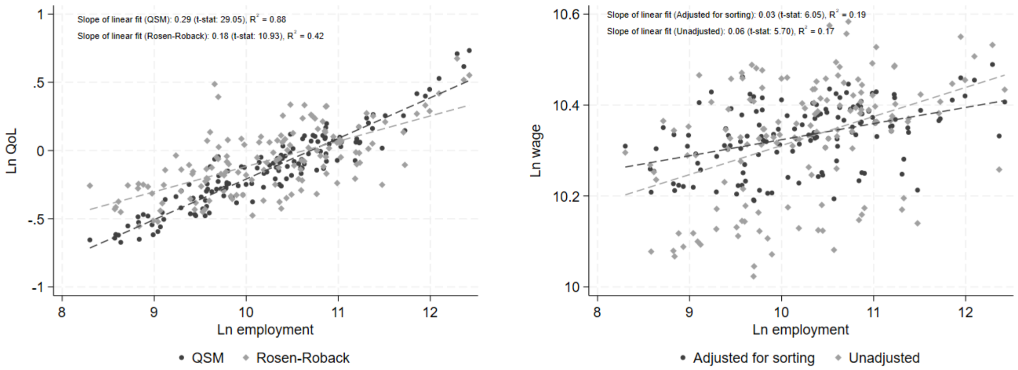
- **First-order explanation:** agglomeration economies
  - sheer magnitude of spillovers in prime locations
  - dense matching of firms, workers, and ideas
  - concentration of very high productivity and revenue per worker (e.g. Manhattan)
- **But firms also need workers**
  - especially high-skill, mobile workers
  - whose location choices increasingly matter
  - firms partly follow workers, not only the other way round
- **Amenities and quality of life increasingly shape worker location**
  - Skilled workers sort into high-amenity cities (Diamond, 2016)

Q: Is there a role for quality of life in urban growth?

Are cities good places to work and live?

- **Urban QoL premium is intuitively plausible**
  - cities in better places grow larger; larger cities offer more **consumption variety**
  - Cities attractive to **young & high-skilled** workers (Couture & Handbury, 2020)
- **Neoclassical quality-of-life literature**
  - higher QoL shows up as **lower real wages** via rent capitalization (Roback, 1982)
  - empirical literature finds **little evidence** of a positive urban QoL premium
- **Quantitative model with spatial frictions**
  - Need to account for idiosyncratic tastes, local ties, tradable goods, local services
  - **Large cities must offer a QoL premium** to attract more workers
  - Downward bias in **larger urban QoL premium** in Rosen-Roback framework
  - Ahlfeldt, Bald, Roth, Seidel (2025)

The urban QoL premium in Germany



- Urban QoL premium > urban wage premium (Ahlfeldt, Bald, Roth, Seidel, 2025)
  - Elasticity of 0.18-0.29 vs. 0.03-0.06
  - Urban QoL premium increased by 20% from 2007 to 2019

What's behind the urban QoL premium?

- Large cities offer high wages and high QoL
  - **Urban pull from the production and consumption side**
  - Both are likely **increasing over time**
- Urban QoL and wage premia are both **descriptive concepts**, no causality
  - Fixed costs in local services imply that **consumption variety** increases with city size
  - **Natural amenities** (climate, mountains, rivers) attract workers (reverse causality)
- QoL may capture amenity value of **insurance against labour demand shocks**
  - Cities trade wage growth against wage risk depending on industry composition
  - Large cities of closer to the efficient frontier (lower risk at same growth)
  - **Diversification and risk pooling** (Zhang, 2026)

From diagnosis to policy

- Spatial inequalities are **not an accident**
  - Increasing returns to density
  - Structural transformation
  - Superstar firms
  - Urban quality of life
- Increase **aggregate productivity** but also increase **spatial inequalities**
  - Places that don't matter" and political polarization
    - Rodriguez-Pose (2018), Autor et al. (2020), etc.
  - Demand for place-based policies (Südekum 2025)
- **Agglomeration is part of the problem and part of the solution**

Q: (How) Can policy shape the spatial growth-equity trade-off?

From spatial efficiency to spatial incidence

- **Traditional spatial equilibrium models (Rosen 1979, Roback 1982)**
  - Perfect labour mobility
  - Spatial equilibrium equalizes utility
  - Regional policies mainly reallocate activity across space
  - Little scope for lasting regional welfare gains
- **Recent quantitative spatial equilibrium models**
  - Heterogeneous preferences (Allen & Arkolakis, 2014, Ahlfeldt et al. 2015)
  - Local ties and moving costs (Caliendo et al., 2019; Greaney et al., 2025)
  - Housing supply matters due to downward-sloping housing demand
  - Regional policies generate **spatial incidence**
- Policy implication
  - Place-based policies need not create spatial misallocation
  - Regional policies should **shift the national efficiency-regional equity frontier**

Spatially balanced agglomeration

- Agglomeration should occur...
  - ...several spatially distributed cities rather in only a few superstar cities
- Create a network of **regional growth poles**
  - Attractive places to live
  - Highly productive places to work
  - Within commuting distance of surrounding regions
- Spatial diversification also helps people in superstar cities
  - Strengthening regional growth poles complements housing supply in superstar cities
  - Sustainable growth in many attractive specialized cities reduces migration pressure
- Goal
  - A connected system of **productive, livable and complementary cities**

Policy principle I: Build on comparative advantage

- Large cities
  - Innovation and entrepreneurship
  - Advanced producer services
  - Universities and knowledge creation
  - Urbanization economies
- Smaller cities
  - Specialized industries and services
  - Localization economies
  - Lower land and housing costs
  - Lower congestion and high livability
- Policy implication
  - **Do not create small versions of Berlin or Munich**
  - Build **highly productive, specialized, mid-sized cities**
  - Duranton & Puga (2001)

Policy principle II: Connect the urban system

- Think about **regional policy and infrastructure policy jointly**
  - High-performance commuter rail & broadband $\Rightarrow$ spillovers to regional centers
- Develop a modern **hierarchy of places**
  - Metropolitan areas generate innovation
  - Regional centres specialize and diffuse growth
- Connect cities through
  - Competitive commuter rail
  - Excellent digital infrastructure
  - Remote work and integrated labour markets
- Cities should **complement** rather than imitate each other

Policy principle III: Consider QoL

- Firms increasingly compete for **people**
- People increasingly choose cities based on **quality of life**
  - Housing affordability
  - Amenities
  - Public services
  - Urban environment
- Regional policy therefore needs to combine
  - Productivity
  - Quality of life
  - Housing
  - Infrastructure
- Livability is no longer a by-product of growth
  - It is an input into regional competitiveness

Conclusion

- **Spatial inequalities are the consequence of powerful agglomeration forces**
  - Density, structural transformation, superstar firms and QoL reinforce each other
- **Agglomeration is part of the solution—not the problem**
  - The challenge is to spread its benefits without sacrificing productivity
- **Spatially balanced agglomeration**
  - A connected network of productive, specialized and livable regional growth poles

References I

- Ahlfeldt, G. M., Albers, T., & Behrens, K. (2022). Prime services and urban growth.
- Ahlfeldt, G. M., Albers, T., & Behrens, K. (2026). A granular spatial model, work in project.
- Ahlfeldt, G. M., Bald, F., Roth, D., & Seidel, T. (2025). The urban quality-of-life premium. Working paper.
- Ahlfeldt, G. M., Heblich, S., Seidel, T., & Yin, F. (2026). Urban productivity and commercial rents. Working paper.
- Ahlfeldt, G. M., & Pietrostefani, E. (2019). The economic effects of density: A synthesis. *Journal of Urban Economics*, 111, 93–107.
- Ahlfeldt, G. M., Redding, S. J., Sturm, D. M., & Wolf, N. (2015). The economics of density: Evidence from the Berlin Wall. *Econometrica*, 83(6), 2127–2189.
- Allen, T., & Arkolakis, C. (2014). Trade and the topography of the spatial economy. *Quarterly Journal of Economics*, 129(3), 1085–1140.
- Autor, D., Dorn, D., Katz, L. F., Patterson, C., & Van Reenen, J. (2020). The fall of the labor share and the rise of superstar firms. *Quarterly Journal of Economics*, 135(2), 645–709.
- Caliendo, L., Dvorkin, M., & Parro, F. (2019). Trade and labor market dynamics: General equilibrium analysis of the China trade shock. *Econometrica*, 87(3), 741–835.

References II

- Chen, W., Novy, D., Perroni, C., & Wong, M. (2025). Urban-biased structural transformation. Working paper.
- Couture, V., & Handbury, J. (2020). Urban revival in America. *Journal of Urban Economics*, 119, 103267.
- Dauth, W., Findeisen, S., Moretti, E., & Suedekum, J. (2022). Matching in cities. *Journal of the European Economic Association*.
- De Loecker, J., Eeckhout, J., & Unger, G. (2020). The rise of market power and the macroeconomic implications. *Quarterly Journal of Economics*, 135(2), 561–644.
- Diamond, R. (2016). The determinants and welfare implications of US workers' diverging location choices. *American Economic Review*, 106(3), 479–524.
- Duranton, G., & Puga, D. (2001). Nursery cities: Urban diversity, process innovation, and the life cycle of products. *American Economic Review*, 91(5), 1454–1477.
- Duranton, G., & Puga, D. (2004). Micro-foundations of urban agglomeration economies. In J. V. Henderson & J.-F. Thisse (Eds.), *Handbook of Regional and Urban Economics*, Vol. 4.
- Eckert, F., Ganapati, S., & Walsh, C. (2025). The geography of new work. Working paper.
- Greaney, B., et al. (2025). Mobility frictions and regional adjustment. Working paper.

References III

- Iammarino, S., Rodríguez-Pose, A., & Storper, M. (2019). Regional inequality in Europe: Evidence, theory and policy implications. *Journal of Economic Geography*, 19(2), 273–298.
- Marshall, A. (1890). *Principles of Economics*. Macmillan.
- Moretti, E. (2013). *The New Geography of Jobs*. Houghton Mifflin Harcourt.
- Roback, J. (1982). Wages, rents, and the quality of life. *Journal of Political Economy*, 90(6), 1257–1278.
- Rodríguez-Pose, A. (2018). The revenge of the places that don't matter. *Cambridge Journal of Regions, Economy and Society*, 11(1), 189–209.
- Südekum, J. (2025). Place-based policies – How to do them and why. *Global Challenges & Regional Science*, 1, 100003.
- Zhang, A. (2026). Urban diversification, growth and labour-market risk. Working paper.